

Retail Sales Analytics Project Report

Executive Summary

This project analyzes the Global Superstore retail dataset to identify trends in sales, profitability, customer behavior, product performance, regional performance, shipping efficiency, and overall business profitability.

The analysis was performed using Python for data cleaning, SQL Server for business analysis through 67 analytical queries, and Power BI to develop seven interactive dashboards that enable executives and managers to make informed business decisions.

Key Findings

1. Executive Overview

Findings

Total Sales: R12.64 Million

Total Profit: R1.47 Million

Profit Margin: 11.61%

Total Orders: 25,040

Unique Customers: 4,870

Technology generated the highest sales and profit.

APAC is the largest sales market.

Sales increased consistently from 2011 to 2014.

Business Recommendations

Continue investing in Technology products.

Maintain strong customer relationships within the APAC market.

Replicate successful sales strategies across lower-performing markets.

Monitor profit margin regularly to improve operational efficiency.

2. Sales Performance Analysis

Findings

Monthly sales increased steadily over the four-year period.

Sales reached their highest level in 2014.

Sales grew by approximately 27% during the strongest year.

Profit followed the same upward trend.

Business Recommendations

Increase inventory before high-demand periods.

Expand successful promotional campaigns.

Forecast demand using historical sales trends.

Monitor Year-over-Year growth monthly.

3. Customer Analysis

Findings

Consumer Segment contributes over half of total sales.

Several customers generate significant losses.

A small number of customers contribute disproportionately to revenue.

Customer Lifetime Value exceeds R12 Million.

Business Recommendations

Reward high-value customers through loyalty programmes.

Investigate customers generating repeated losses.

Improve customer retention strategies.

Target Corporate and Home Office segments with specialised marketing campaigns.

4. Product Analysis

Findings

Technology products dominate overall sales.

Phones and Copiers are among the highest-performing subcategories.

Several products consistently generate losses.

Product profitability varies significantly across categories.

Business Recommendations

Increase stock of high-performing products.

Review pricing of loss-making products.

Discontinue consistently unprofitable products.

Expand successful product categories.

5. Regional Analysis

Findings

APAC records the highest sales.

Central region produces the highest revenue.

New York is among the highest-performing cities.

Regional profitability differs considerably.

Business Recommendations

Allocate more marketing resources to high-performing regions.

Develop improvement strategies for weaker regions.

Expand operations within profitable cities.
Investigate causes of poor regional performance.

6. Discount and Profitability Analysis

Findings

Overall profit margin is 11.61%.
Thousands of products and orders generate losses.
Furniture records the lowest profit margin.
Technology maintains the strongest profitability.

Business Recommendations

Reduce excessive discounting.
Review pricing strategy for Furniture.
Remove consistently loss-making products.
Improve profitability through supplier negotiations.

7. Shipping & Logistics Analysis

Findings

Standard Class accounts for the majority of shipments.
Same Day shipping has the highest average shipping cost.
Standard Class generates the highest overall profit.
Shipping cost differs considerably between shipping modes.

Business Recommendations

Encourage customers to choose Standard Class where appropriate.
Optimise Same Day shipping costs.
Improve logistics planning to reduce transportation expenses.
Monitor shipping efficiency continuously.

Overall Business Insights

The business demonstrates consistent growth in sales and profit throughout the analysis period.
Technology remains the strongest-performing product category, while APAC is the leading market.
Although profitability is healthy, several products and customers generate recurring losses, presenting opportunities for optimisation.
Shipping operations are generally efficient, but premium shipping methods increase operational costs.

Overall Recommendations

Continue expanding Technology product offerings.

Focus investment on high-performing markets such as APAC.

Strengthen customer loyalty programmes.

Review pricing strategies for low-margin products.

Eliminate or redesign consistently loss-making products.

Improve logistics efficiency to reduce shipping costs.

Use predictive analytics to forecast future demand.

Develop executive dashboards for continuous performance monitoring.

Regularly review customer profitability before offering discounts.

Use SQL and Power BI for continuous business intelligence reporting.